

## 1-Task Title: Game of Chomp

### Code:

```
import random
print("Which side you want to take? Player 1 or Player 2\nPlayer",end=' ')
x=int(input())
n=20
if x==1:
    y=False
    a=2
else:
    y=True
    a=1
while True:
    if y is True:
        if n<=4:
            m=n
        else:
            m=random.randint(1,4)
        n-=m
        print(f"Player {a} takes {m} balls. {n} balls are left")
        y=False
        if n==0:
            print(f"Player {a} is win")
            break
    else:
        ball=0
        while ball>4 or ball<=0:
            ball=int(input("Enter your choice: "))
        n-=ball
        print(f"Player {x} takes {ball} balls. {n} balls are left")
        y=True
        if n==0:
            print(f"Player {x} is win")
            break
```

### Output:

```
Which side you want to take? Player 1 or Player 2
Player 2
Player 1 takes 2 balls. 18 balls are left
Enter your choice: 4
Player 2 takes 4 balls. 14 balls are left
Player 1 takes 1 balls. 13 balls are left
Enter your choice: 3
Player 2 takes 3 balls. 10 balls are left
Player 1 takes 2 balls. 8 balls are left
Enter your choice: 1
Player 2 takes 1 balls. 7 balls are left
Player 1 takes 2 balls. 5 balls are left
Enter your choice: 1
Player 2 takes 1 balls. 4 balls are left
Player 1 takes 4 balls. 0 balls are left
Player 1 is win
Press any key to continue . . . |
```

## Discussion:

The game is very interesting to me as there is a logical process to how can first player can win the match easily. If second player also knows the logic and if first gives any wrong move then second player also can win. In this game one can takes ball at most 4 balls in each move and total ball is 20. If any player unable to take any ball then the opposite player will win the game.

## 2-Task Title: Weird Algorithm

### Code:

```
n=int(input("Enter the value: "))
print(n,end=' ')
while n!=1:
    if n%2==0:
        n/=2
        print(int(n),end=' ')
    else:
        n*=3
        n+=1
        print(int(n),end=' ')
print("\n")
```

### Output:

```
Enter the value: 1111
1111 3334 1667 5002 2501 7504 3752 1876 938 469 1408 704 352 176 88 44 22 11 34
17 52 26 13 40 20 10 5 16 8 4 2 1

Press any key to continue . . . |
```

## Discussion:

The task name is Weird Algorithm which comes from the 'CSES' problem set. Here I have to do the operation till the number is not equal to 1. The operations are for even numbers I have to divide that number by 2, for odd numbers I have to multiply that number by 3 and then add 1 with it. This operations will continuously be run until the program reaches to 1. As it is not proven for all numbers, so it is a conjecture.